Supplementary Materials
BMEF-D-26-00169

Supplementary Results

Table S1: **Demographic characteristics and prepared clinical inputs of ADNI.** Data are mean ± sample s.d., except counts. Demographic P values use one-way ANOVA (age, weight) or a chi-square test (sex). Clinical summaries combine observed and diagnostic-group-conditioned filled scores; their group-comparison P values are omitted because the distributions are not independent clinical evidence. Source missingness is reported in Table S4. CDR: Clinical Dementia Rating; MMSE: Mini-Mental State Examination.

	CN	**EMCI**	**LMCI**	**MCI**	**AD**	**P**
n	577	291	152	520	317	
Age (years)	72.55 ± 6.65	71.62 ± 7.51	72.44 ± 7.69	73.20 ± 7.51	74.90 ± 7.99	<0.001
Weight (kg)	73.20 ± 24.13	80.77 ± 16.67	77.15 ± 16.64	72.90 ± 22.33	73.21 ± 18.64	0.0023
Sex (M/F)	252/325	165/126	80/72	310/210	186/131	<0.001
MMSE	29.40 ± 0.70	27.50 ± 0.50	24.47 ± 0.50	26.83 ± 1.24	20.84 ± 1.97	–
GDSCALE	0.24 ± 0.73	1.00 ± 0.00	3.00 ± 0.00	1.81 ± 0.99	4.14 ± 1.62	–
Global CDR	0.00 ± 0.02	0.15 ± 0.23	0.50 ± 0.00	0.42 ± 0.19	0.95 ± 0.15	–
FAQ	0.52 ± 0.53	3.11 ± 2.42	8.83 ± 1.54	5.33 ± 1.51	20.20 ± 2.92	–
NPI-Q	0.01 ± 0.19	1.55 ± 2.17	4.00 ± 0.00	2.92 ± 0.50	4.98 ± 0.65	–

Note: CN = Cognitively Normal; EMCI = Early MCI; LMCI = Late MCI; MCI = the remaining MCI participants not assigned to EMCI or LMCI (MCI-U in the text); AD = Alzheimer's dementia. Zero standard deviations reflect constant prepared scores, including diagnosis-conditioned fills; they do not establish absence of clinical variability. The original CN weight entries span 0–227 kg and are retained without exclusions or replacements; their clinical validity and the meaning of zero entries remain unresolved.

The historical OASIS-3 external cohort comprised 419 PUP-processed AV45 participants (334 CN, 85 AD); the 117 FDG participants without the corresponding processing were not included. The age distributions differed between diagnostic groups (Table S2). Reported clinical summaries do not establish which complete prompt fields were used in every external run or how unavailable

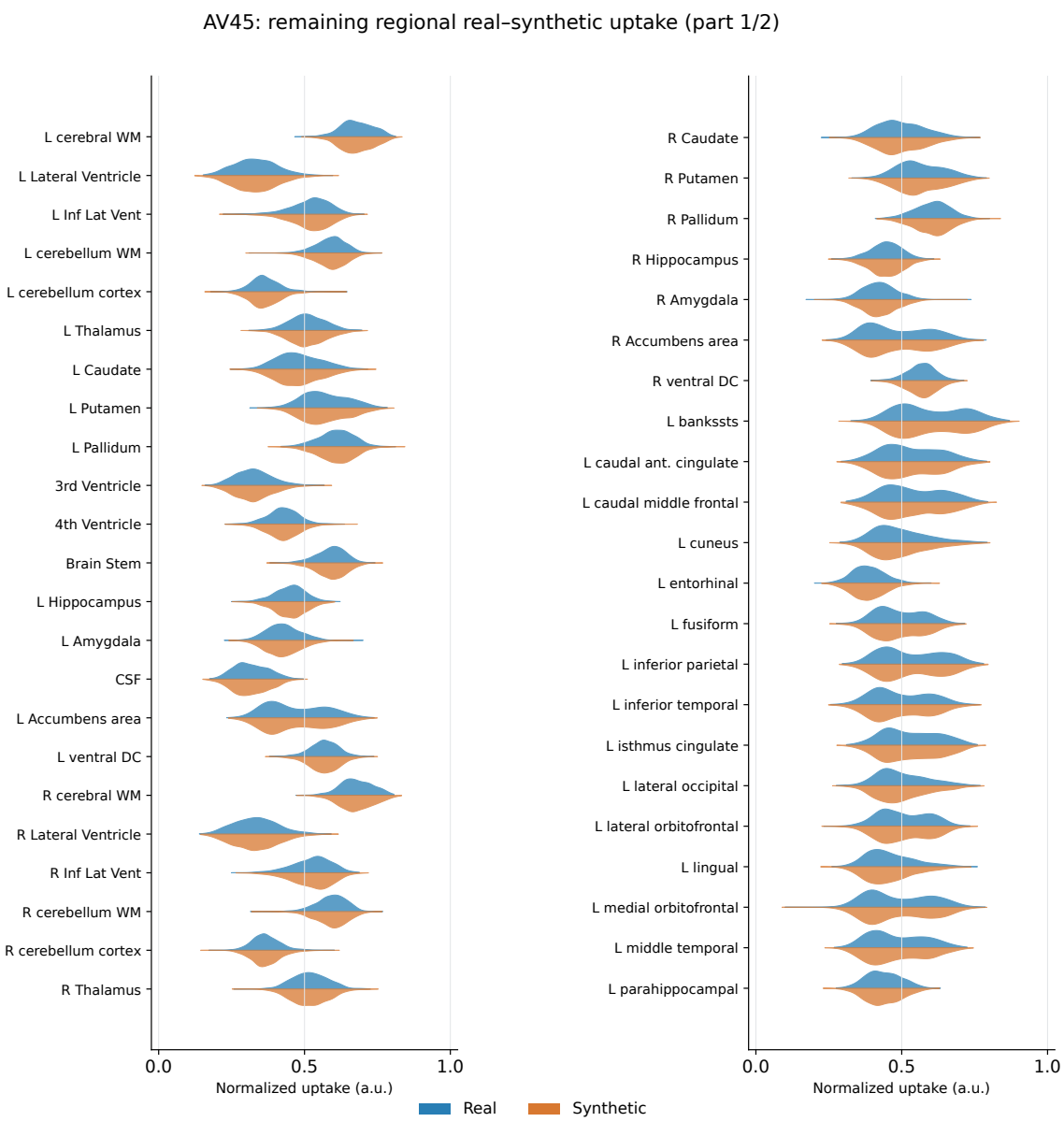


Figure S1: **Comparison of real and synthetic AV45 uptake across the remaining brain regions (Part 1 of 2).** Values represent mean regional intensities from per-image min–max-normalized PET images (arbitrary units), rather than SUVR.

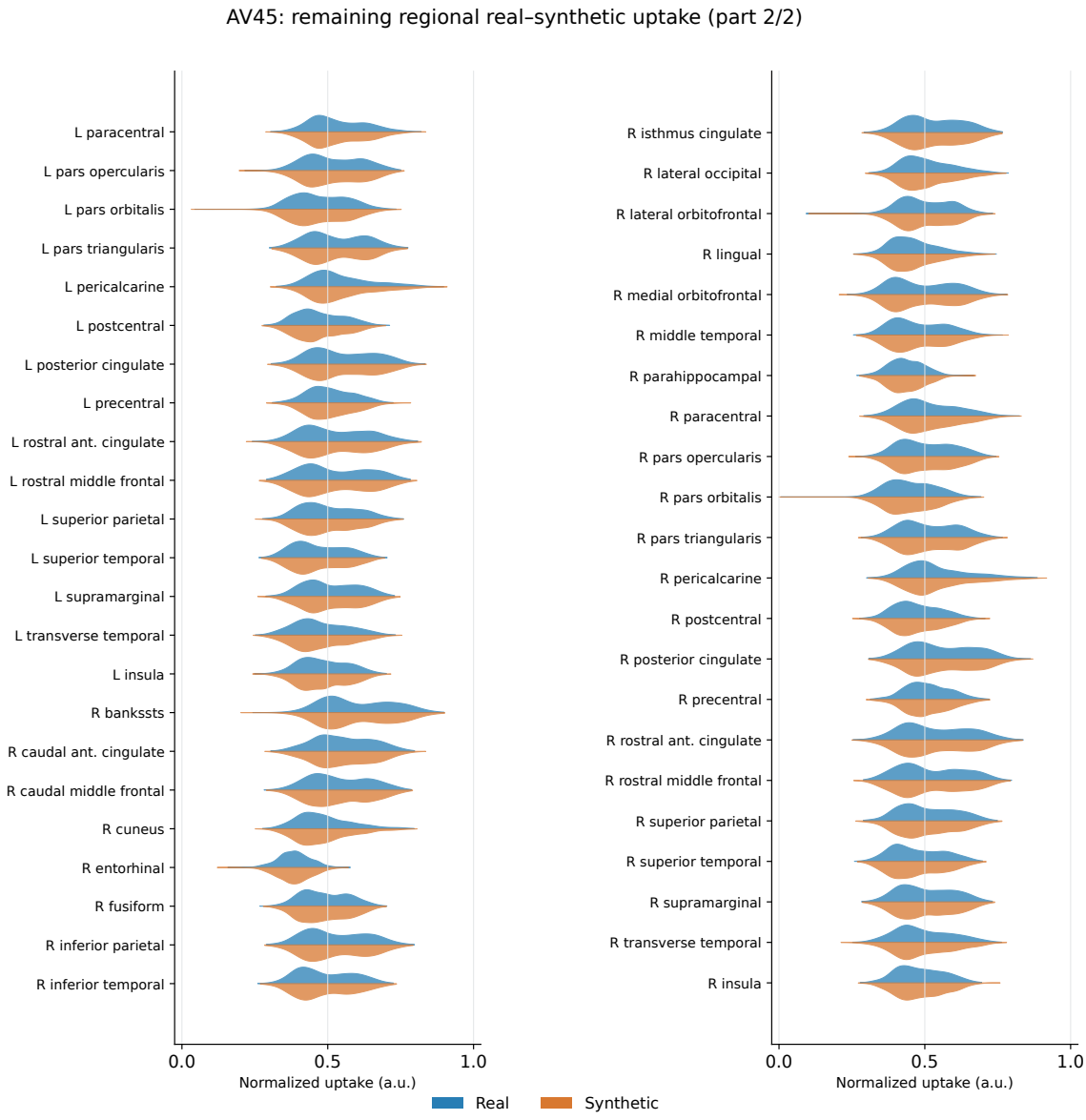


Figure S2: **Comparison of real and synthetic AV45 uptake across the remaining brain regions (Part 2 of 2).** Values represent mean regional intensities from per-image min–max-normalized PET images (arbitrary units), rather than SUVR.

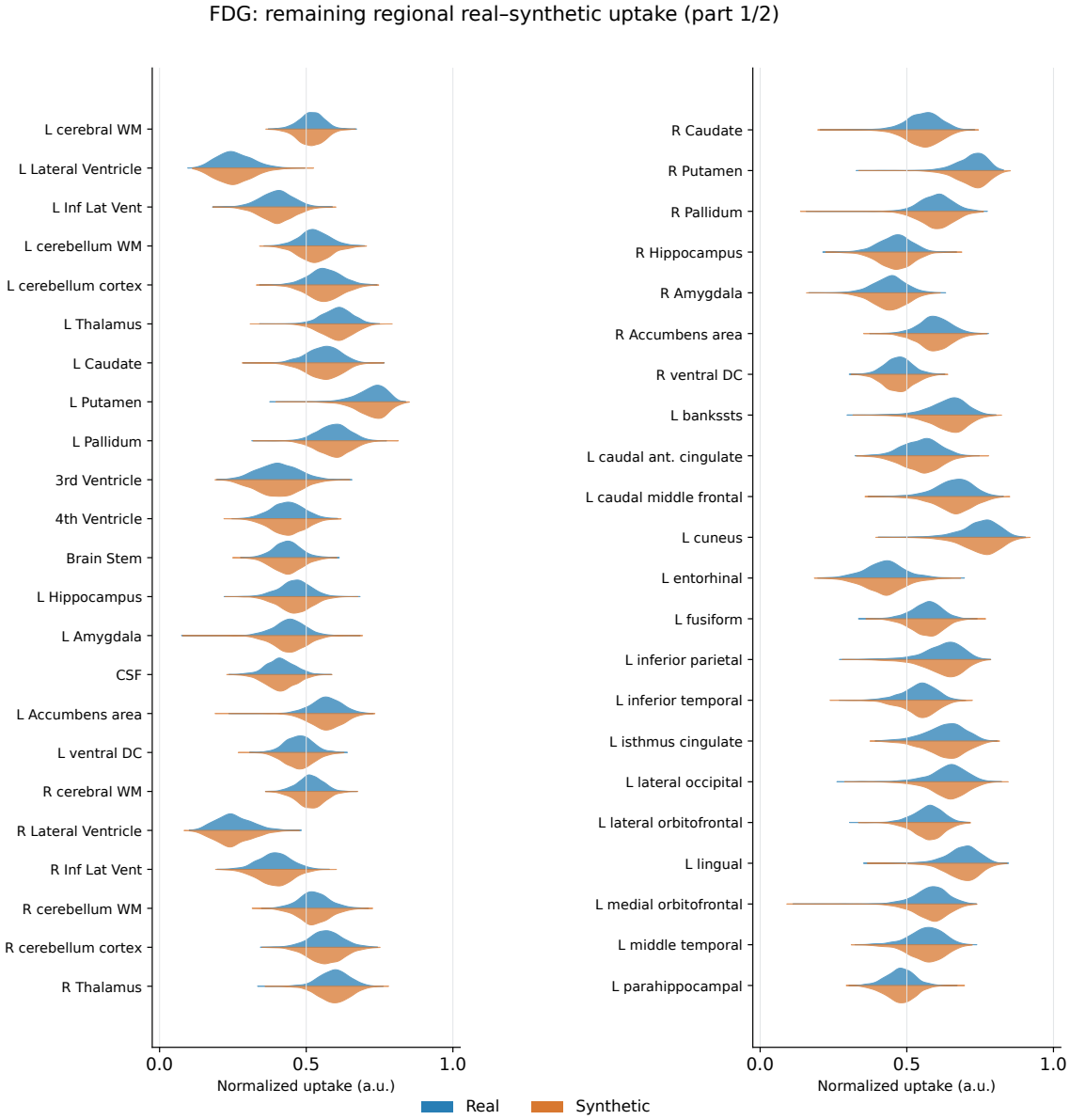


Figure S3: **Comparison of real and synthetic FDG uptake across the remaining brain regions (Part 1 of 2).** Values represent mean regional intensities from per-image min–max-normalized PET images (arbitrary units), rather than SUVR.

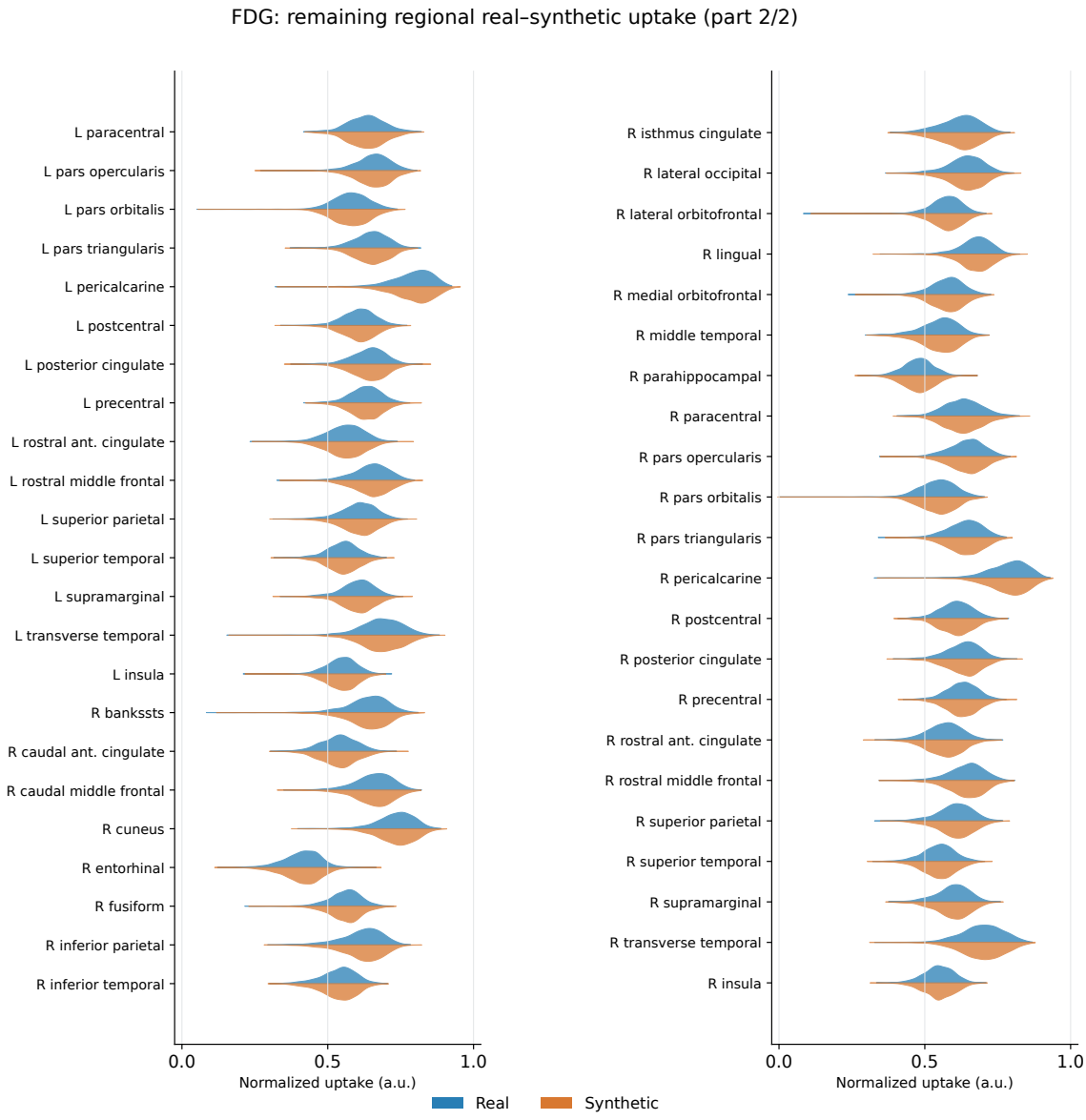


Figure S4: **Comparison of real and synthetic FDG uptake across the remaining brain regions (Part 2 of 2).** Values represent mean regional intensities from per-image min–max-normalized PET images (arbitrary units), rather than SUVR.

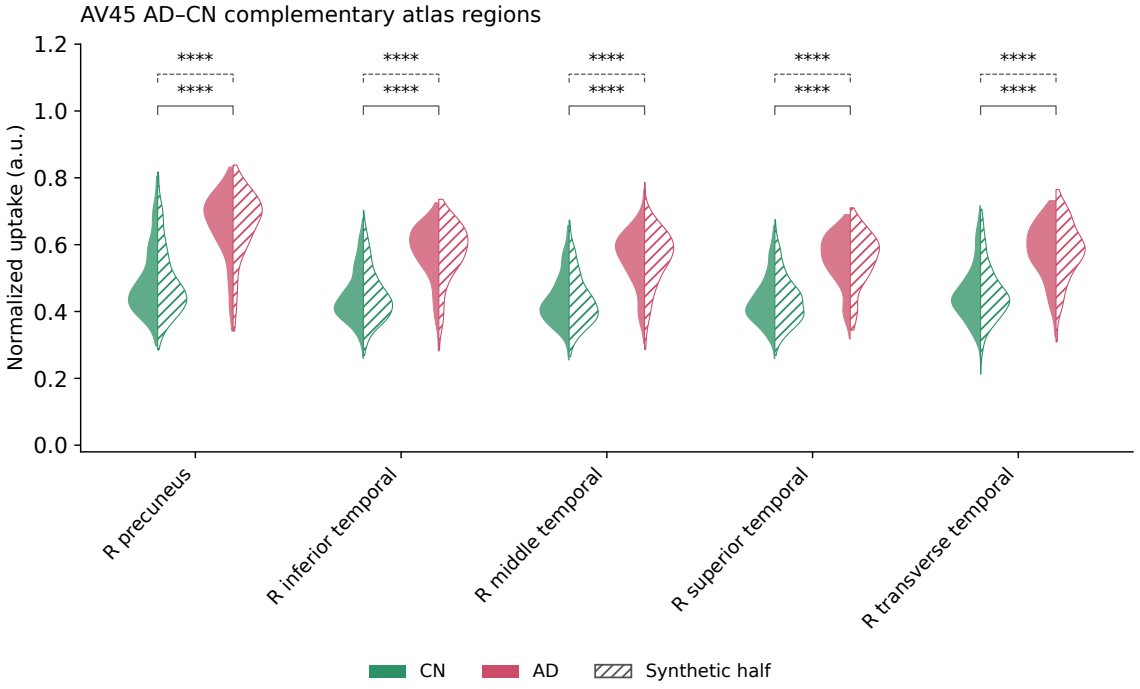


Figure S5: **Validation of disease-related AV45 uptake patterns across the remaining brain regions.** Violin plots display mean normalized regional uptake (arbitrary units) in Alzheimer's disease (AD, red) and cognitively normal (CN, green) subjects for selected brain regions. Each region contains paired half-violins: the left half (solid fill) represents real PET data, and the right half (dashed fill) represents MRI-synthetic PET data, with consistent color coding between modalities. Solid lines above the violins indicate statistical significance for real PET, while dashed lines indicate significance for synthetic PET. Uptake values were derived from per-image min–max-normalized PET images without reference-region normalization.

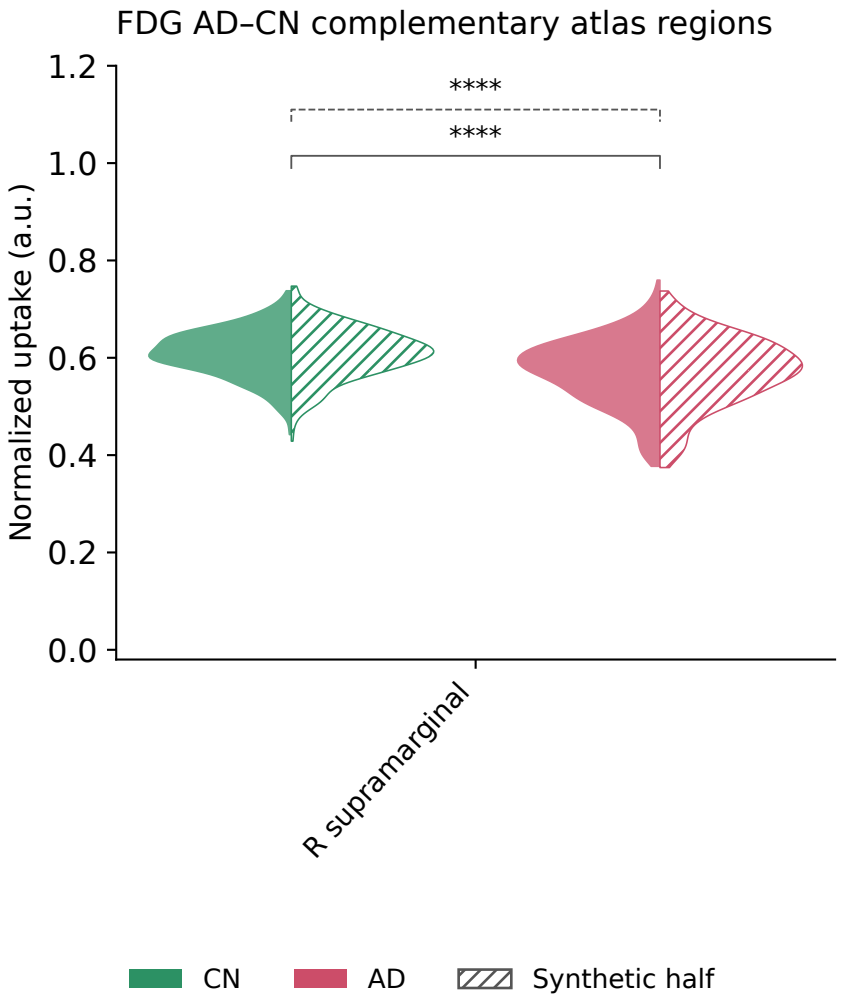


Figure S6: **Validation of disease-related FDG uptake patterns across the remaining brain regions.** Violin plots display mean normalized regional uptake (arbitrary units) in Alzheimer's disease (AD, red) and cognitively normal (CN, green) subjects for selected brain regions. Each region contains paired half-violins: the left half (solid fill) represents real PET data, and the right half (dashed fill) represents MRI-synthetic PET data, with consistent color coding between modalities. Solid lines above the violins indicate statistical significance for real PET, while dashed lines indicate significance for synthetic PET. Uptake values were derived from per-image min–max-normalized PET images without reference-region normalization.

fields were handled. The external results are restricted to AV45 synthesis in CN/AD and do not provide FDG or MCI external evidence.

Table S2: **Demographic characteristics of OASIS-3.**

	CN	**AD**	**P**
n	334	85	
Age (years)	66.33 ± 8.38	72.76 ± 6.96	<0.001
Sex (M/F)	143/191	37/48	>0.05
MMSE score	29.24 ± 0.90	23.21 ± 3.51	<0.001
Global CDR	0.01 ± 0.08	0.73 ± 0.59	<0.001

Note: CN = Cognitively Normal; AD = Alzheimer's dementia.

Table S3: **Historical core software dependencies and reported versions for DIReCT++.** This retained environment list describes the historical configuration; reconstructed training and the timing run used separately recorded environments. In particular, the RTX 4090 timing run used PyTorch 2.5.1+cu121, as stated in Methods. The table is not a single environment specification for every experiment.

Package	**Version**	**Package**	**Version**
Python	3.10	PyTorch	2.6.0+cu118
torchvision	0.21.0+cu118	torchaudio	2.6.0+cu118
CUDA	11.8	cuDNN	9.1.0
einops	0.8.1	transformers	4.40.2
tokenizers	0.19.1	safetensors	0.5.3
xformers	0.0.29.post3	faiss	1.9.0
monai	1.5.1	numpy	2.2.4
scipy	1.15.2	pandas	2.2.3
scikit-learn	1.6.1	matplotlib	3.10.1
seaborn	0.13.2	opencv-python	4.12.0.88
nibabel	5.3.2	tensorboard	2.19.0
fsspec	2024.6.1	huggingface-hub	0.30.1
protobuf	6.30.2	triton	3.2.0

Supplementary Methods: conditioning prompts and adapter analysis

The following is the historical clinical prompt template, with bracketed field placeholders instead of participant identifiers or invented measurements. Values use the historical formatting (rounded age and scores, one decimal for weight and CDR). Its use does not validate the filled values as clinical observations.

> Subject is a [age]-year-old [sex] with a weight of [weight] kg. The global Clinical Dementia Rating (CDR) score, which assesses dementia severity (0: no dementia to 3: severe dementia), is [global_cdr]. The Mini-Mental State Examination (MMSE) score, assessing cognitive function (0: severe impairment to 30: normal), is [mmse_score]. The Geriatric Depression Scale (GDS) score, screening depression (0: no depression to 15: severe depression), is [gds_score].

Table S4: **Missing clinical scores in the source records before historical filling.** Each cell gives missing count / group size (percentage); missingness is defined relative to the first source CSV record selected for each participant, not all ADNI visits.

Score	CN	EMCI	LMCI	MCI-U	AD
MMSE	427/577	291/291	152/152	289/520	234/317
	(74.0%)	(100.0%)	(100.0%)	(55.6%)	(73.8%)
GDSCALE	429/577	291/291	152/152	289/520	234/317
	(74.4%)	(100.0%)	(100.0%)	(55.6%)	(73.8%)
Global CDR	427/577	291/291	152/152	289/520	234/317
	(74.0%)	(100.0%)	(100.0%)	(55.6%)	(73.8%)
FAQ	561/577	167/291	151/152	489/520	306/317
	(97.2%)	(57.4%)	(99.3%)	(94.0%)	(96.5%)
NPI-Q	563/577	167/291	152/152	491/520	306/317
	(97.6%)	(57.4%)	(100.0%)	(94.4%)	(96.5%)

Source records contain 1,751 observed and 7,534 missing score cells across 1,857 participants and five scores. All available scores were retained. Historical filling used diagnostic-group rules: MMSE values were drawn from group-specific integer sets; GDSCALE and NPI-Q used group-specific fixed values; FAQ used fixed values or group-specific integer sets/ranges; and CDR used fixed values or a group-specific Bernoulli choice for EMCI/MCI-U. These are simulated inputs, not clinical observations or group-mean imputation. Their use can encode diagnostic-group information. The original random states were not recovered, and no re-imputation was performed. Age, sex and weight have no blank entries in the prepared table, which does not establish the clinical validity of every entry. The clinical source row's Image ID does not identify the actual MRI/PET acquisition used by the model.

Table S5: **Reproducibility settings and scope of the revised analyses.**

Component	Recorded settings and limitations
Reconstructed generation	1,857 subjects; fixed two-fold partitions of 929/928, split seed 42. Full, TracerOnly, TabularMLP and Single did not use a fixed training seed; deterministic repeated-training behavior is not claimed. Epoch 165 is the principal comparison endpoint; MRI-only Swin-UNETR uses its prespecified epoch 200.
Classification	Task-specific subject-level stratified three-fold splits (seed 42); 15% stratified internal validation within each outer training fold. Fit seed $42 + 100 \times$ task index+fold, with task index AD/CN=0, MCI/CN=1, EMCI/LMCI=2 and folds 1–3. Training lasts 200 epochs with validation every two epochs; checkpoint selected by validation ACC; outer test evaluated once after reloading.
Classification cohorts	AD/CN: 470 (165 AD, 305 CN); MCI/CN: 741 (436 EMCI+LMCI, 305 CN); EMCI/LMCI: 436 (287 EMCI, 149 LMCI). MCI-U excluded. The task-specific partitions are shared across input conditions.
Source identity	The historical classification Syn-PET inputs are distinct from the reconstructed generation comparisons; a direct checkpoint binding between these image families was not established. Existing classification inputs were not regenerated.
Clinical source	Clinical rows selected by first CSV record per participant; five score fields contain diagnosis-conditioned fills. Original filling random states and complete actual acquisition mappings were not recovered.

The Functional Activities Questionnaire (FAQ) score, assessing daily activity impairment (0: no impairment to 30: severe impairment), is [faq_score]. The Neuropsychiatric Inventory Questionnaire (NPI-Q) Total Score, assessing neuropsychiatric symptom burden (0: no symptoms to higher scores indicating greater burden), is [npi_q_score].

FDG tracer prompt.

FDG PET is a functional brain imaging technique that visualizes the dynamic changes in glucose metabolism, directly linked to neuronal energy demands and synaptic activity. It serves as a tool to assess functional connectivity and energy utilization across brain regions. Areas with decreased metabolic activity, such as those affected by neurodegenerative diseases, should exhibit reduced signal intensity. High-intensity metabolic hotspots in gray matter (e.g., the cerebral cortex and basal ganglia) are key markers of neuronal activity.

AV45 tracer prompt.

AV45 PET is a molecular imaging technique that highlights the static distribution of amyloid-beta plaques, a critical pathological marker of Alzheimer's disease. This imaging modality provides a spatial map of amyloid deposition in cortical regions (e.g., the temporal, parietal, and frontal lobes) and can distinguish amyloid-positive areas from amyloid-negative white matter regions. The primary focus is on identifying amyloid deposition patterns to assess disease progression and pathological burden.

The tracer texts are fixed model inputs; their statements about pathology or progression are not evidence that synthetic PET establishes a biological diagnosis or predicts progression.

Table S6: **Adapter-level sensitivity analysis of the cross-tracer soft target.** Descriptive means across two folds, using 1,856 QC-filtered participants. Only training-fold PET features were used for fitting; held-out features were evaluated after optimization. No cross-term means removal of the penalty ($\lambda_{\text{cross}} = 0$), not a target of zero. No generator was retrained or evaluated in this analysis.

Setting	Training alignment		Held-out alignment		Cross-tracer
	FDG	AV45	FDG	AV45	similarity
$\tau = 0.70$	0.9437	0.9410	0.9436	0.9410	0.7032
$\tau = 0.82$	0.9687	0.9661	0.9687	0.9661	0.8214
$\tau = 0.90$	0.9838	0.9813	0.9838	0.9812	0.9018
$\tau = 0.98$	0.9948	0.9923	0.9948	0.9923	0.9807
No cross-term	0.9952	0.9927	0.9952	0.9926	0.9897

Table S7: **Regional normalized uptake agreement and descriptive regression for the 16 representative tracer–ROI combinations in Figure 4A,B.** Overall cohorts from the author-designated historical ROI CSVs; N can vary by ROI.

Tracer	ROI (ID)	N	Mean diff. [95% CI]	MAE	CCC	ICC(A,1)	BA bias [95% LoA]	p	q	OLS slope [95% CI]	Intercept [95% CI]
			0.00052				0.0005			1.0127	-0.0064
AV45	L precuneus (1025)	1365	[-0.00056, 0.00159]	0.0163	0.9863	0.9863	[-0.0391, 0.0401]	0.3445	0.9786	[1.0038, 1.0215]	[-0.0114, -0.0014]
			0.00021				0.0002			1.0008	-0.0002
AV45	R precuneus (2025)	1365	[-0.00079, 0.00120]	0.0150	0.9887	0.9887	[-0.0366, 0.0370]	0.6846	0.9786	[0.9928, 1.0088]	[-0.0047, 0.0043]
			0.00033				0.0003			1.0037	-0.0012
AV45	L frontalpole (1032)	1365	[-0.00072, 0.00138]	0.0157	0.9808	0.9808	[-0.0385, 0.0392]	0.5384	0.9786	[0.9932, 1.0143]	[-0.0055, 0.0032]
			0.00039				0.0004			1.0007	0.0001
AV45	R frontalpole (2032)	1364	[-0.00068, 0.00147]	0.0160	0.9815	0.9815	[-0.0392, 0.0400]	0.4746	0.9786	[0.9904, 1.0110]	[-0.0042, 0.0045]
			-0.00009				-0.0001			0.9794	0.0078
AV45	L temporalpole (1033)	1365	[-0.00117, 0.00099]	0.0162	0.9591	0.9591	[-0.0399, 0.0398]	0.8769	0.9786	[0.9640, 0.9947]	[0.0019, 0.0137]
			0.00031				0.0003			1.0087	-0.0032
AV45	R temporalpole (2033)	1365	[-0.00076, 0.00138]	0.0163	0.9644	0.9644	[-0.0392, 0.0399]	0.5667	0.9786	[0.9942, 1.0232]	[-0.0092, 0.0027]
			-0.00037				-0.0004			1.0002	-0.0005
AV45	L superiorfrontal (1028)	1365	[-0.00146, 0.00073]	0.0165	0.9806	0.9807	[-0.0407, 0.0400]	0.5102	0.9786	[0.9896, 1.0107]	[-0.0058, 0.0049]
			0.00063				0.0006			0.9970	0.0021
AV45	R superiorfrontal (2028)	1365	[-0.00041, 0.00168]	0.0157	0.9822	0.9822	[-0.0380, 0.0393]	0.2354	0.9786	[0.9869, 1.0071]	[-0.0030, 0.0073]
			-0.00003				-0.0000			0.9844	0.0108
FDG	L precuneus (1025)	1598	[-0.00100, 0.00093]	0.0156	0.9573	0.9573	[-0.0386, 0.0385]	0.9433	0.9774	[0.9699, 0.9990]	[0.0006, 0.0209]
			0.00010				0.0001			0.9983	0.0013
FDG	R precuneus (2025)	1598	[-0.00086, 0.00105]	0.0156	0.9598	0.9598	[-0.0380, 0.0382]	0.8428	0.9774	[0.9841, 1.0125]	[-0.0086, 0.0111]
			0.00024				0.0002			1.0039	-0.0019
FDG	L frontalpole (1032)	1598	[-0.00074, 0.00122]	0.0160	0.9586	0.9586	[-0.0389, 0.0394]	0.6273	0.9410	[0.9895, 1.0184]	[-0.0100, 0.0061]
			-0.00032				-0.0003			1.0041	-0.0026
FDG	R frontalpole (2032)	1597	[-0.00130, 0.00065]	0.0159	0.9634	0.9634	[-0.0394, 0.0387]	0.5160	0.9410	[0.9906, 1.0177]	[-0.0103, 0.0050]
			-0.00034				-0.0003			0.9934	0.0023
FDG	L temporalpole (1033)	1598	[-0.00132, 0.00063]	0.0159	0.9430	0.9430	[-0.0392, 0.0385]	0.4901	0.9410	[0.9764, 1.0105]	[-0.0046, 0.0091]
			0.00068				0.0007			0.9902	0.0047
FDG	R temporalpole (2033)	1598	[-0.00030, 0.00166]	0.0161	0.9449	0.9449	[-0.0386, 0.0400]	0.1754	0.9410	[0.9736, 1.0069]	[-0.0022, 0.0116]
			0.00093				0.0009			1.0101	-0.0052
FDG	L superiorfrontal (1028)	1598	[-0.00006, 0.00191]	0.0159	0.9438	0.9438	[-0.0385, 0.0403]	0.0657	0.9410	[0.9931, 1.0271]	[-0.0156, 0.0051]
			0.00081				0.0008			1.0020	-0.0004
FDG	R superiorfrontal (2028)	1598	[-0.00017, 0.00178]	0.0158	0.9465	0.9465	[-0.0382, 0.0398]	0.1055	0.9410	[0.9855, 1.0185]	[-0.0104, 0.0096]

Differences and Bland–Altman (BA) bias are synthetic minus real uptake (arbitrary units after per-image min–max normalization). Mean-difference confidence intervals and individual BA limits of agreement are distinct. CCC is Lin's concordance correlation and ICC(A,1) is a two-way absolute-agreement single-measure coefficient. Paired two-sided t-test p values are raw; BH-FDR q values use all 98 ROIs within each tracer's overall cohort. OLS regresses synthetic on real uptake and gives unadjusted descriptive 95% CIs. Non-significant differences and slopes close to one do not establish equivalence, clinical SUV/SUVR calibration or interchangeable biological measurements.

Table S8: **AD–CN regional effect sizes and BH-FDR-adjusted comparisons for Figure 4C–F.** Cohen's d is AD minus CN divided by the pooled within-group sample s.d.

Panel	Tracer	ROI (ID)	N_{AD}	N_{CN}	Real d	Syn d	Real q	Syn q
C	AV45	L precuneus (1025)	179	511	1.6783	1.6437	7.73×10^{-66}	1.25×10^{-63}
C	AV45	L inferiortemporal (1009)	179	511	1.6831	1.6337	4.68×10^{-66}	4.74×10^{-63}
C	AV45	L middletemporal (1015)	179	511	1.6482	1.6236	4.67×10^{-64}	1.72×10^{-62}
C	AV45	L superiortemporal (1030)	179	511	1.5780	1.5434	5.64×10^{-60}	8.51×10^{-58}
C	AV45	L transversetemporal (1034)	179	511	1.5133	1.5288	2.84×10^{-56}	5.61×10^{-57}
D	FDG	L isthmuscingulate (1010)	303	371	-1.2531	-1.1774	2.69×10^{-48}	2.10×10^{-43}
D	FDG	R isthmuscingulate (2010)	303	371	-1.3972	-1.3201	1.18×10^{-57}	1.94×10^{-52}
D	FDG	L supramarginal (1031)	303	371	-0.4848	-0.4308	2.14×10^{-9}	1.10×10^{-7}
D	FDG	Left Hippocampus (17)	302	371	-0.6401	-0.6042	4.27×10^{-15}	1.27×10^{-13}
D	FDG	Right Hippocampus (53)	303	371	-0.7518	-0.7189	4.59×10^{-20}	1.54×10^{-18}
E	AV45	Left VentralDC (28)	179	511	-0.1050	-0.0933	0.2368	0.2858
E	AV45	Right VentralDC (60)	179	511	-0.1211	-0.0986	0.1729	0.2618
E	AV45	Right Hippocampus (53)	179	511	-0.0068	0.0006	0.9373	0.9949
E	AV45	Left Cerebellum White Matter (7)	179	511	-0.0745	-0.1010	0.3996	0.2529
E	AV45	Left Hippocampus (17)	179	511	0.0701	0.1232	0.4241	0.1667
F	FDG	Left Thalamus (10)	303	371	-0.0097	0.0582	0.9008	0.4667
F	FDG	Right Thalamus (49)	303	371	-0.0668	-0.0530	0.3964	0.4991
F	FDG	L parsorbitalis (1019)	303	371	-0.1294	-0.0913	0.1035	0.2506
F	FDG	L lateraloccipital (1011)	303	371	-0.1239	-0.0978	0.1171	0.2229
F	FDG	R lingual (2013)	303	371	-0.0493	-0.0435	0.5303	0.5744

Separate two-sided Student's t-tests compare AD and CN for each tracer and PET type. BH-FDR correction uses all 98 ROIs within each of four tracer-by-PET-type families, rather than just the displayed regions. Positive d indicates higher uptake in AD. All values are derived from min–max-normalized regional intensities, not SUV/SUVR. Preserved significance or similar effect sizes do not establish clinical equivalence.

Table S9: Whole-volume OOF generation metrics (mean ± sample s.d.)

Model	Tracer	N	SSIM	PSNR	MSE	MAE
Full	AV45	1365	0.884408 ± 0.0405641	24.9664 ± 2.26749	0.00372914 ± 0.00332507	0.0309555 ± 0.0110338
Full	FDG	1598	0.921129 ± 0.0416829	27.223 ± 2.14503	0.00224384 ± 0.00268009	0.024129 ± 0.00961375
ClinicalUNet	AV45	1365	0.860412 ± 0.0374027	24.637 ± 2.02806	0.00388392 ± 0.00241146	0.0324286 ± 0.00954057
ClinicalUNet	FDG	1598	0.900311 ± 0.0356273	26.6116 ± 1.89074	0.00247114 ± 0.00196941	0.0258502 ± 0.00818077
SwinUNETR	AV45	1365	0.874091 ± 0.0369388	24.5068 ± 2.11433	0.00403421 ± 0.00249859	0.0320631 ± 0.00986262
SwinUNETR	FDG	1598	0.918317 ± 0.0360297	26.8441 ± 1.9234	0.00234143 ± 0.00181013	0.0241961 ± 0.00799078
TabularMLP	AV45	1365	0.874855 ± 0.0407899	25.1009 ± 2.28897	0.00363223 ± 0.00321331	0.0308469 ± 0.0110175
TabularMLP	FDG	1598	0.912262 ± 0.0417162	27.2919 ± 2.12402	0.00220694 ± 0.0026887	0.0241971 ± 0.00947495
TracerOnly	AV45	1365	0.885748 ± 0.0408475	24.9695 ± 2.27491	0.00373732 ± 0.00335201	0.0308555 ± 0.0111714
TracerOnly	FDG	1598	0.926504 ± 0.0407948	27.2745 ± 2.14158	0.00221882 ± 0.00270009	0.0236521 ± 0.00949478
Single	AV45	1365	0.873767 ± 0.041007	24.9848 ± 2.21914	0.00369036 ± 0.00320827	0.031269 ± 0.0107387
Single	FDG	1598	0.926933 ± 0.0400237	27.0223 ± 2.11834	0.00234305 ± 0.00287526	0.0239414 ± 0.00968223

Mean plus/minus sample SD. Whole-volume frozen OOF; FDG1598, AV451365. Full165, ClinicalUNet165, Swin200 MRI-only, Single165, TabularMLP165, TracerOnly165. SSIM unitless; PSNR dB; MSE squared and MAE unsquared normalized-intensity arbitrary units. Independent training realizations; reconstructed runs, not historical Figure2.

Table S10: Paired whole-volume generation comparisons

model	tracer	metric	n	mean_delta	ci_low	ci_high	cohen_dz	Full better pct	p_raw	p_holm
ClinicalUNet	fdg	ssim	1598	0.0208181	0.0197791	0.0218175	0.990561	89.8623	1.7603e-239	1.40824e-238
ClinicalUNet	fdg	psnr	1598	0.611381	0.5664	0.656514	0.648354	76.5957	6.43853e-124	3.86312e-123
ClinicalUNet	fdg	mse	1598	-0.0002273	-0.000275411	-0.000158772	-0.189496	76.5957	6.04086e-14	1.31586e-13
ClinicalUNet	fdg	mae	1598	-0.00172122	-0.0019156	-0.00151429	-0.417989	75.4068	6.9045e-58	3.45225e-57
ClinicalUNet	av45	ssim	1365	0.0239956	0.0227111	0.025253	0.978795	87.1062	2.29545e-201	1.60681e-200
ClinicalUNet	av45	psnr	1365	0.329409	0.249281	0.40985	0.220001	62.0513	9.68848e-16	3.87539e-15
ClinicalUNet	av45	mse	1365	-0.000154787	-0.000257116	-3.72771e-05	-0.0752304	62.0513	0.00551979	0.00551979
ClinicalUNet	av45	mae	1365	-0.00147312	-0.00185782	-0.00109689	-0.206517	63.7363	4.3862e-14	1.31586e-13
SwinUNETR	fdg	ssim	1598	0.00281191	0.00204043	0.00354844	0.175086	69.2741	3.77993e-12	1.88996e-11
SwinUNETR	fdg	psnr	1598	0.378929	0.334097	0.423834	0.40333	69.0238	2.67057e-54	1.8694e-53
SwinUNETR	fdg	mse	1598	-9.75975e-05	-0.000146119	-3.08259e-05	-0.0822636	69.0238	0.00102917	0.00205835
SwinUNETR	fdg	mae	1598	-6.70702e-05	-0.000253042	0.000134417	-0.0166045	55.8198	0.506936	0.506936
SwinUNETR	av45	ssim	1365	0.0103163	0.00965209	0.0109524	0.822615	83.4432	2.31981e-155	1.85585e-154
SwinUNETR	av45	psnr	1365	0.459673	0.393978	0.526625	0.375353	66.8132	5.23792e-41	3.14275e-40
SwinUNETR	av45	mse	1365	-0.000305079	-0.000395415	-0.000193256	-0.160805	66.8132	3.58919e-09	1.07676e-08
SwinUNETR	av45	mae	1365	-0.00110762	-0.0014185	-0.000784845	-0.18539	61.4652	1.11886e-11	4.47543e-11
TabularMLP	fdg	ssim	1598	0.00886656	0.00812186	0.00959853	0.578166	75.8448	3.44323e-102	2.75458e-101
TabularMLP	fdg	psnr	1598	-0.0689397	-0.108799	-0.029079	-0.083771	47.3091	0.000830676	0.00425048
TabularMLP	fdg	mse	1598	3.68949e-05	1.37346e-05	5.99572e-05	0.0775476	47.3091	0.00196924	0.00787697
TabularMLP	fdg	mae	1598	-6.81443e-05	-0.000213963	8.04426e-05	-0.0227372	53.3166	0.363529	0.727058
TabularMLP	av45	ssim	1365	0.00955263	0.00863504	0.0104854	0.54034	74.2125	5.41378e-78	3.78964e-77
TabularMLP	av45	psnr	1365	-0.134409	-0.210528	-0.0561215	-0.0918656	46.8132	0.000708413	0.00425048
TabularMLP	av45	mse	1365	9.69028e-05	1.94112e-05	0.000173411	0.0677817	46.8132	0.0123868	0.0371605
TabularMLP	av45	mae	1365	0.000108651	-0.000232867	0.000435387	0.0173217	51.1355	0.522301	0.727058
TracerOnly	fdg	ssim	1598	-0.00537535	-0.00590994	-0.00484902	-0.47433	26.9086	1.79244e-72	1.43395e-71
TracerOnly	fdg	psnr	1598	-0.0514788	-0.084006	-0.0198543	-0.0783114	48.8736	0.0017767	0.00888351
TracerOnly	fdg	mse	1598	2.5016e-05	5.74477e-06	4.40603e-05	0.0629846	48.8736	0.0119058	0.0476234
TracerOnly	fdg	mae	1598	0.000476913	0.000360989	0.000595372	0.197824	42.3029	4.84122e-15	3.38885e-14
TracerOnly	av45	ssim	1365	-0.0013401	-0.00186153	-0.000811475	-0.132833	44.1026	1.03289e-06	6.19734e-06
TracerOnly	av45	psnr	1365	-0.00307791	-0.0547682	0.0476535	-0.0032182	47.9853	0.905373	1
TracerOnly	av45	mse	1365	-8.18237e-06	-5.70224e-05	4.02036e-05	-0.00891382	47.9853	0.741957	1
TracerOnly	av45	mae	1365	0.000100047	-0.000115014	0.000313809	0.024782	46.8132	0.360042	1

model	tracer	metric	n	mean_delta	ci_low	ci_high	cohen_dz	Full better pct	p_raw	p_holm
Single	fdg	ssim	1598	-0.00580378	-0.00642246	-0.00517901	-0.434763	34.2303	4.22015e-62	2.95411e-61
Single	fdg	psnr	1598	0.200636	0.159293	0.242418	0.237103	60.5131	8.98722e-21	5.39233e-20
Single	fdg	mse	1598	-9.92098e-05	-0.000126504	-7.21876e-05	-0.177651	60.5131	1.84961e-12	9.24804e-12
Single	fdg	mae	1598	0.000187601	3.76336e-05	0.00034206	0.0596374	47.3717	0.0172416	0.0527503
Single	av45	ssim	1365	0.0106401	0.00985376	0.0114433	0.707251	78.1685	2.31929e-122	1.85543e-121
Single	av45	psnr	1365	-0.0183712	-0.0770958	0.0388809	-0.0169039	49.8168	0.532384	0.532384
Single	av45	mse	1365	3.87787e-05	-1.71053e-05	9.51783e-05	0.0366412	49.8168	0.176043	0.352086
Single	av45	mae	1365	-0.000313478	-0.000560844	-6.84978e-05	-0.0671769	54.8718	0.0131876	0.0527503

Delta=Full minus comparator; positive favors Full for SSIM/PSNR and negative for MSE/MAE. Cohen dz signed.10000 paired resamples within frozen folds, seed42; percentile95% CIs unadjusted. Two-sided paired t; separate Holm8-test family per comparator (2tracers x4metrics). ClinicalUNet statistics reused; TracerOnly t/dz reused and missing bootstrap supplemented. CIs do not estimate training/seed variability. MRI-only Swin inputs and endpoint differ.

Table S11: Mask-restricted quality of frozen two-fold OOF PET predictions.

Model	Tracer	N	SSIM ↑	PSNR (dB) ↑	MSE ↓	MAE ↓
DIReCT++ Full (e165)	FDG	1597	0.91844 ± 0.04026	23.954 ± 2.271	0.004763 ± 0.004028	0.05257 ± 0.01921
DIReCT++ Full (e165)	AV45	1364	0.86999 ± 0.06048	21.803 ± 2.602	0.007959 ± 0.005469	0.06845 ± 0.02394
Clinical Regression 3D U-Net (e165)	FDG	1597	0.90208 ± 0.04256	23.373 ± 2.019	0.005262 ± 0.003889	0.05509 ± 0.01795
Clinical Regression 3D U-Net (e165)	AV45	1364	0.86098 ± 0.05868	21.529 ± 2.363	0.008237 ± 0.005320	0.06976 ± 0.02276
Swin-UNETR (e200; MRI only)	FDG	1597	0.89237 ± 0.03938	23.565 ± 2.013	0.004997 ± 0.003300	0.05404 ± 0.01672
Swin-UNETR (e200; MRI only)	AV45	1364	0.83615 ± 0.05878	21.273 ± 2.429	0.008791 ± 0.005652	0.07228 ± 0.02363

Values are mean ± sample SD across subjects. SSIM is unitless; PSNR is in dB; MSE and MAE use squared and unsquared [0,1] normalized-uptake units, respectively. All models use byte-identical frozen target PET arrays and identical subject-specific masks. The mask is the frozen SynthSeg nonzero-label union, including cortex, white matter, subcortical structures, cerebellum, brainstem, ventricles and CSF; it is not a pure parenchymal mask. SSIM uses the existing E1 3D Gaussian implementation (kernel 7, sigma 1.5, data range 1) and only centers whose entire 7×7×7 window lies inside the mask. The frozen mask QC exclusion removes one subject from each tracer; no new exclusions or spatial/intensity transformations were applied. Full and Clinical U-Net use epoch 165; MRI-only Swin-UNETR uses the prespecified final epoch 200. All predictions are frozen two-fold OOF outputs.

Table S12: Paired mask-restricted PET image-quality comparisons.

Comparison	Tracer	Metric	N	Mean Δ	95% bootstrap CI	t	Cohen d_z	Full better (%)	Raw p	Holm p
A: Full − Clinical U-Net	FDG	SSIM	1597	0.01636	[0.01577, 0.01699]	51.477	1.288	95.7	$< 10^{-300}$	$< 10^{-300}$
A: Full − Clinical U-Net	FDG	PSNR	1597	0.581	[0.529, 0.634]	21.474	0.537	72.3	4.55e-90	3.18e-89
A: Full − Clinical U-Net	FDG	MSE	1597	-0.000499	[-0.000572, -0.000425]	-13.116	-0.328	72.3	2.13e-37	1.28e-36
A: Full − Clinical U-Net	FDG	MAE	1597	-0.00253	[-0.00295, -0.00210]	-11.670	-0.292	66.9	2.96e-30	1.18e-29
A: Full − Clinical U-Net	AV45	SSIM	1364	0.00901	[0.00754, 0.01051]	11.881	0.322	69.0	4.79e-31	2.39e-30
A: Full − Clinical U-Net	AV45	PSNR	1364	0.273	[0.178, 0.370]	5.585	0.151	57.8	2.82e-08	8.45e-08
A: Full − Clinical U-Net	AV45	MSE	1364	-0.000278	[-0.000488, -0.000066]	-2.609	-0.071	57.8	0.00917	0.00962
A: Full − Clinical U-Net	AV45	MAE	1364	-0.00131	[-0.00223, -0.00040]	-2.824	-0.076	56.4	0.00481	0.00962
B: Full − Swin-UNETR	FDG	SSIM	1597	0.02607	[0.02553, 0.02663]	92.200	2.307	99.0	$< 10^{-300}$	$< 10^{-300}$
B: Full − Swin-UNETR	FDG	PSNR	1597	0.389	[0.337, 0.443]	14.239	0.356	66.8	2.11e-43	1.26e-42
B: Full − Swin-UNETR	FDG	MSE	1597	-0.000233	[-0.000311, -0.000151]	-5.733	-0.143	66.8	1.18e-08	1.18e-08
B: Full − Swin-UNETR	FDG	MAE	1597	-0.00148	[-0.00191, -0.00105]	-6.681	-0.167	63.3	3.26e-11	6.52e-11
B: Full − Swin-UNETR	AV45	SSIM	1364	0.03383	[0.03274, 0.03493]	59.128	1.601	95.2	$< 10^{-300}$	$< 10^{-300}$
B: Full − Swin-UNETR	AV45	PSNR	1364	0.530	[0.451, 0.609]	13.216	0.358	65.1	1.3e-37	6.52e-37
B: Full − Swin-UNETR	AV45	MSE	1364	-0.000832	[-0.001003, -0.000658]	-9.563	-0.259	65.1	5.15e-21	1.55e-20
B: Full − Swin-UNETR	AV45	MAE	1364	-0.00383	[-0.00459, -0.00309]	-10.003	-0.271	64.1	8.81e-23	3.52e-22

Δ = DIReCT++ Full minus comparator on exactly matched subject, tracer and held-out fold. Positive differences favor Full for SSIM/PSNR; negative differences favor Full for MSE/MAE. Cohen's dz uses this signed difference divided by its sample SD. Full-better proportions use strict metric-specific improvement. Two-sided paired t-tests; 95% percentile bootstrap CIs use 10,000 paired resamples, seed 42, stratified by held-out fold with original fold sizes retained. Holm correction is applied separately to comparison A and comparison B, with eight tests (2 tracers × 4 metrics) in each family; CIs are unadjusted. Uncertainty concerns the frozen OOF subjects, not retraining/seed variability. Statistical significance is not clinical equivalence. The two comparisons differ in information and endpoint: A uses matched MRI/clinical/tracer information at e165; B compares Full e165 with MRI-only Swin-UNETR e200. A displayed $p < 10^{-300}$ denotes numerical underflow of the computed p value, not an exactly zero probability.

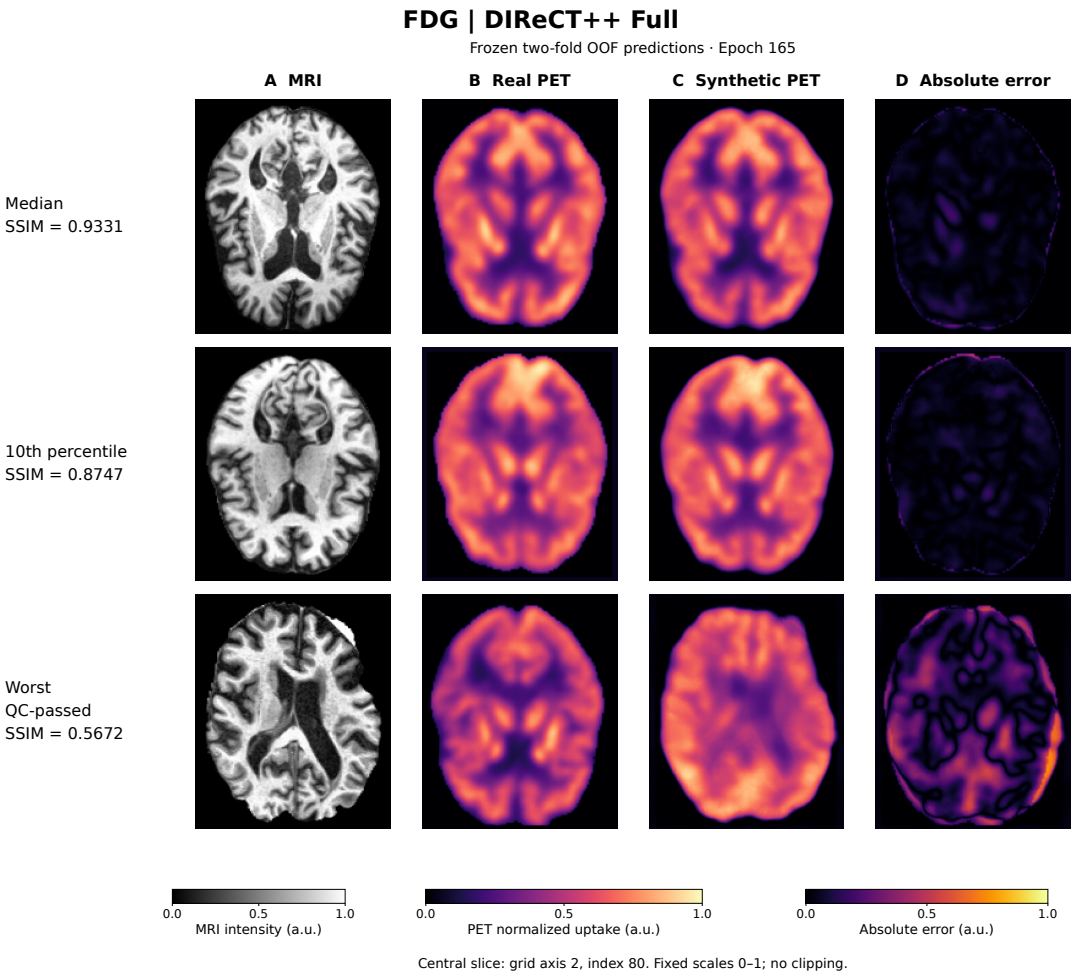


Figure S7: **FDG synthesis in median, 10th-percentile and worst QC-passed cases.** Rows follow the frozen whole-volume SSIM selection; columns show MRI, real PET, synthetic PET and absolute-error maps. PET intensities and errors use the displayed [0,1] normalized-intensity scales. Selection is within the existing QC-passed set, without new exclusions or visual replacement; rankings use whole-volume SSIM rather than masked SSIM.

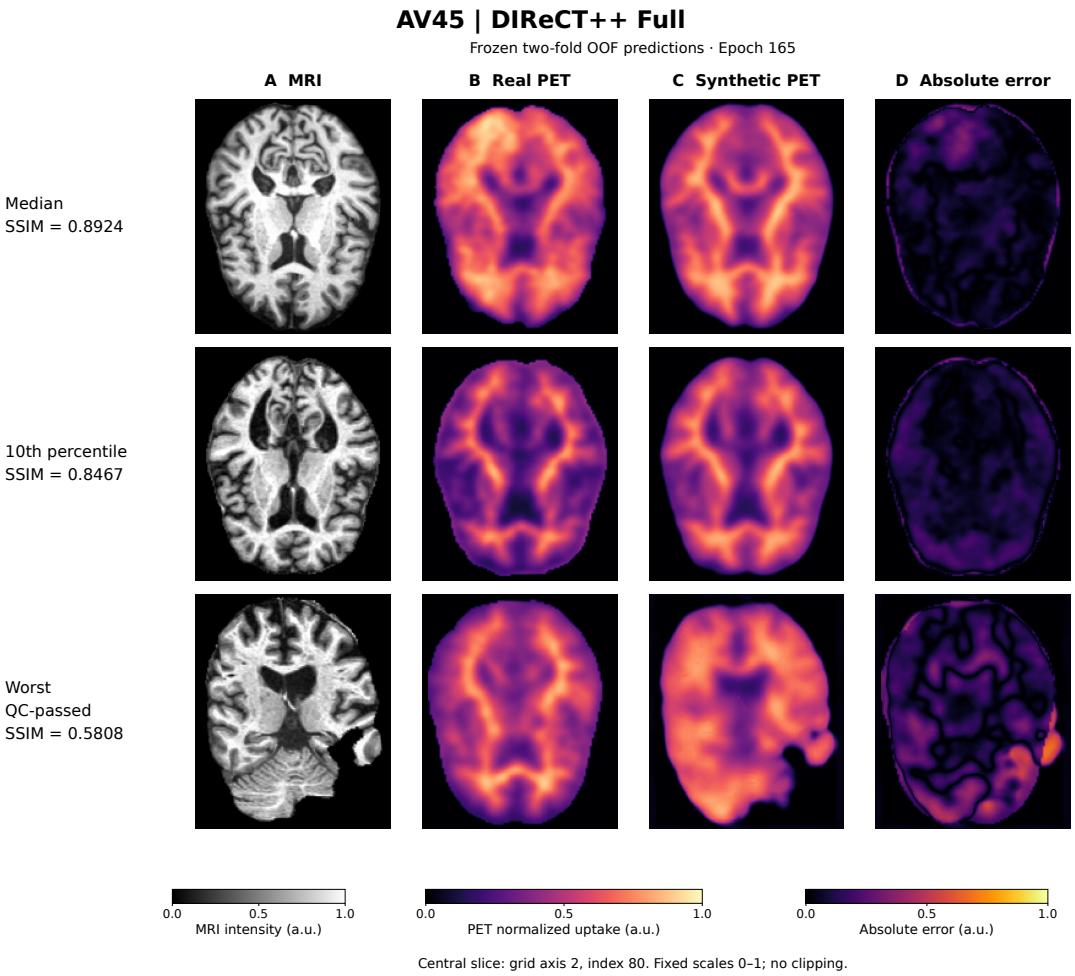


Figure S8: **AV45 synthesis in median, 10th-percentile and worst QC-passed cases.** Rows follow the frozen whole-volume SSIM selection; columns show MRI, real PET, synthetic PET and absolute-error maps. PET intensities and errors use the displayed [0,1] normalized-intensity scales. Selection is within the existing QC-passed set, without new exclusions or visual replacement; rankings use whole-volume SSIM rather than masked SSIM.

Table S13: Three-fold classification results: mean ± sample s.d.; [conditional 95% CI].

Input	ACC	AUC	SEN	SPE	Balanced ACC
AD/CN					
MRI	79.56 ± 6.20 [75.95, 82.97]	87.02 ± 5.44 [83.51, 90.15]	75.15 ± 10.01 [68.48, 81.21]	81.96 ± 6.90 [77.70, 85.91]	78.56 ± 6.59 [74.63, 82.24]
MRI+Real FDG	89.36 ± 0.72 [86.39, 91.92]	94.28 ± 0.55 [91.77, 96.51]	81.21 ± 6.39 [75.15, 87.27]	93.78 ± 4.41 [90.85, 96.40]	87.50 ± 1.06 [84.07, 90.69]
MRI+Syn FDG	88.72 ± 3.31 [85.93, 91.48]	94.40 ± 0.89 [91.96, 96.58]	80.00 ± 12.73 [73.94, 86.06]	93.45 ± 3.95 [90.51, 96.07]	86.73 ± 5.21 [83.35, 89.97]
MRI+Real AV45	85.53 ± 2.67 [82.34, 88.51]	92.54 ± 2.53 [89.91, 94.92]	74.55 ± 1.82 [67.88, 81.21]	91.47 ± 4.86 [88.19, 94.42]	83.01 ± 1.77 [79.27, 86.58]
MRI+Syn AV45	84.04 ± 2.80 [80.84, 87.43]	91.81 ± 3.54 [89.02, 94.23]	81.21 ± 2.10 [75.15, 86.68]	85.56 ± 4.90 [81.62, 89.49]	83.39 ± 2.00 [79.87, 86.91]
MRI+Real both	90.64 ± 2.59 [87.87, 93.18]	94.38 ± 0.90 [91.84, 96.61]	84.24 ± 2.78 [78.18, 89.70]	94.09 ± 2.62 [91.43, 96.40]	89.17 ± 2.60 [85.92, 92.14]
MRI+Syn both	88.08 ± 2.06 [85.09, 90.85]	93.82 ± 3.21 [91.18, 96.07]	86.06 ± 5.84 [80.61, 90.91]	89.18 ± 0.93 [85.57, 92.46]	87.62 ± 2.90 [84.29, 90.67]
Clinical-only	100.00 ± 0.00 [100.00, 100.00]	100.00 ± 0.00 [100.00, 100.00]	100.00 ± 0.00 [100.00, 100.00]	100.00 ± 0.00 [100.00, 100.00]	100.00 ± 0.00 [100.00, 100.00]
MRI+Clinical	94.89 ± 3.54 [92.77, 96.81]	99.17 ± 0.96 [98.50, 99.66]	92.12 ± 4.20 [87.88, 95.76]	96.39 ± 3.44 [94.10, 98.36]	94.26 ± 3.65 [91.90, 96.43]
EMCI+LMCI/CN					
MRI	67.48 ± 0.23 [64.37, 70.58]	71.59 ± 1.80 [67.83, 75.25]	79.57 ± 11.66 [75.89, 83.23]	50.12 ± 17.10 [44.56, 55.38]	64.85 ± 2.73 [61.57, 68.12]
MRI+Real FDG	61.27 ± 2.70 [57.76, 64.78]	64.82 ± 2.79 [60.74, 68.77]	63.30 ± 3.89 [58.69, 67.69]	58.35 ± 1.73 [52.79, 63.90]	60.83 ± 2.48 [57.14, 64.49]
MRI+Syn FDG	62.48 ± 3.25 [59.11, 65.86]	65.77 ± 5.56 [61.81, 69.81]	67.00 ± 12.26 [62.64, 71.13]	56.10 ± 14.64 [50.82, 61.35]	61.55 ± 3.28 [58.05, 65.11]
MRI+Real AV45	66.80 ± 2.10 [63.56, 69.91]	72.00 ± 0.79 [68.23, 75.75]	75.45 ± 8.66 [71.54, 79.35]	54.43 ± 11.27 [48.87, 59.69]	64.94 ± 2.42 [61.54, 68.28]
MRI+Syn AV45	65.99 ± 1.21 [62.75, 69.23]	71.52 ± 1.85 [67.73, 75.10]	73.17 ± 6.51 [69.04, 77.30]	55.76 ± 11.84 [50.49, 61.32]	64.47 ± 2.73 [61.05, 67.90]
MRI+Real both	63.70 ± 3.54 [60.19, 67.21]	69.15 ± 2.42 [65.13, 72.90]	65.60 ± 9.91 [61.24, 69.96]	60.97 ± 8.42 [55.41, 66.21]	63.29 ± 2.81 [59.76, 66.79]
MRI+Syn both	65.45 ± 1.64 [62.48, 68.56]	69.46 ± 1.34 [65.60, 73.18]	79.33 ± 12.80 [75.65, 83.00]	45.53 ± 14.60 [39.95, 51.08]	62.43 ± 0.94 [59.14, 65.71]
Clinical-only	94.87 ± 2.60 [93.25, 96.36]	98.93 ± 0.70 [98.14, 99.56]	93.59 ± 4.17 [91.29, 95.87]	96.72 ± 1.14 [94.43, 98.68]	95.15 ± 2.27 [93.62, 96.59]

Input	ACC	AUC	SEN	SPE	Balanced ACC
MRI+Clinical	85.16 ± 8.58 [82.73, 87.45]	93.07 ± 6.10 [91.14, 94.77]	88.51 ± 12.11 [85.52, 91.26]	80.35 ± 18.35 [76.08, 84.28]	84.43 ± 9.22 [81.82, 86.84]
EMCI/LMCI					
MRI	64.90 ± 3.09 [61.92, 67.89]	62.81 ± 2.50 [57.39, 68.38]	17.40 ± 12.64 [12.03, 23.43]	89.53 ± 6.57 [85.70, 93.01]	53.46 ± 4.63 [50.12, 56.96]
MRI+Real FDG	64.21 ± 4.93 [60.30, 68.11]	61.66 ± 3.35 [56.04, 67.09]	36.86 ± 16.63 [29.48, 44.26]	78.34 ± 15.69 [73.80, 82.87]	57.60 ± 1.88 [53.23, 61.87]
MRI+Syn FDG	65.36 ± 3.46 [61.47, 69.26]	63.02 ± 1.14 [57.49, 68.72]	32.84 ± 5.47 [25.46, 40.88]	82.20 ± 6.97 [77.67, 86.39]	57.52 ± 2.15 [53.15, 62.08]
MRI+Real AV45	65.82 ± 3.65 [62.14, 69.49]	60.62 ± 2.37 [55.03, 66.28]	29.50 ± 9.10 [22.16, 36.87]	84.66 ± 1.30 [80.15, 88.83]	57.08 ± 4.92 [52.84, 61.28]
MRI+Syn AV45	65.37 ± 5.56 [61.47, 69.27]	62.12 ± 6.88 [56.59, 67.70]	36.19 ± 6.89 [28.76, 43.61]	80.46 ± 11.73 [75.92, 84.97]	58.32 ± 2.46 [53.92, 62.83]
MRI+Real both	64.90 ± 4.88 [60.78, 68.81]	63.04 ± 1.25 [57.58, 68.56]	41.05 ± 19.06 [33.66, 48.48]	77.31 ± 13.00 [72.42, 81.85]	59.18 ± 5.52 [54.89, 63.59]
MRI+Syn both	65.82 ± 2.06 [61.69, 70.17]	66.33 ± 0.54 [61.14, 71.76]	43.03 ± 10.87 [34.99, 51.10]	77.71 ± 8.82 [72.83, 82.25]	60.37 ± 1.12 [55.87, 65.19]
Clinical-only	99.08 ± 0.40 [98.16, 99.77]	99.94 ± 0.08 [99.79, 100.00]	98.67 ± 2.31 [96.67, 100.00]	99.31 ± 0.60 [98.26, 100.00]	98.99 ± 0.85 [97.81, 99.83]
MRI+Clinical	85.32 ± 13.80 [82.34, 88.53]	89.60 ± 9.08 [86.11, 93.02]	85.90 ± 12.00 [80.50, 91.28]	85.07 ± 18.35 [81.25, 88.89]	85.49 ± 12.24 [82.16, 88.89]

All metrics are percentages; s.d. is in percentage points. CIs use 5,000 paired fold-by-class bootstrap resamples of frozen predictions and three-fold equal weighting. Intervals are marginal, unadjusted and conditional on the fixed models and splits. Clinical-only AD/CN's zero empirical errors yield a degenerate [100,100] bootstrap interval, not a population guarantee. Clinical inputs include diagnosis-conditioned fills. Sample sizes are AD/CN=470, EMCI+LMCI/CN=741 and EMCI/LMCI=436; MCI-U is excluded.

Table S14: Paired imaging classification differences

task	target	reference	metric	delta_pp	ci_low_pp	ci_high_pp
ad_cn	mri_syn_both	mri	ACC	8.51843	4.47901	12.5606
ad_cn	mri_syn_both	mri	AUC	6.80203	3.38872	10.2184
ad_cn	mri_syn_both	mri	SEN	10.9091	3.63636	18.7879
ad_cn	mri_syn_both	mri	SPE	7.21866	2.31023	12.1369
ad_cn	mri_syn_both	mri	balanced_ACC	9.06388	4.68182	13.457
ad_cn	mri_syn_both	mri_real_both	ACC	-2.55185	-5.31194	-0.00272198
ad_cn	mri_syn_both	mri_real_both	AUC	-0.55129	-2.11379	0.976434
ad_cn	mri_syn_both	mri_real_both	SEN	1.81818	-3.0303	6.66667
ad_cn	mri_syn_both	mri_real_both	SPE	-4.90843	-8.18935	-1.95431
ad_cn	mri_syn_both	mri_real_both	balanced_ACC	-1.54513	-4.34698	1.11062
mci_cn	mri_syn_both	mri	ACC	-2.02429	-5.66802	1.61943
mci_cn	mri_syn_both	mri	AUC	-2.12981	-6.29816	2.09665
mci_cn	mri_syn_both	mri	SEN	-0.239332	-4.60404	4.1301
mci_cn	mri_syn_both	mri	SPE	-4.59458	-11.1499	1.93174
mci_cn	mri_syn_both	mri	balanced_ACC	-2.41695	-6.28258	1.42342
mci_cn	mri_syn_both	mri_real_both	ACC	1.75439	-1.88934	5.39811
mci_cn	mri_syn_both	mri_real_both	AUC	0.308348	-3.14901	3.81391
mci_cn	mri_syn_both	mri_real_both	SEN	13.727	9.14344	18.3059
mci_cn	mri_syn_both	mri_real_both	SPE	-15.4436	-21.3454	-9.55154
mci_cn	mri_syn_both	mri_real_both	balanced_ACC	-0.858316	-4.5701	3.0065
emci_lmci	mri_syn_both	mri	ACC	0.91954	-3.44355	5.50626
emci_lmci	mri_syn_both	mri	AUC	3.52145	-2.42365	9.50467
emci_lmci	mri_syn_both	mri	SEN	25.6327	16.9248	33.7415
emci_lmci	mri_syn_both	mri	SPE	-11.8165	-17.0651	-6.5826
emci_lmci	mri_syn_both	mri	balanced_ACC	6.90807	2.01983	11.9828
emci_lmci	mri_syn_both	mri_real_both	ACC	0.91954	-3.66714	5.50937
emci_lmci	mri_syn_both	mri_real_both	AUC	3.29387	-2.11555	8.53926
emci_lmci	mri_syn_both	mri_real_both	SEN	1.98639	-6.13639	10.0548
emci_lmci	mri_syn_both	mri_real_both	SPE	0.398392	-4.83187	5.965
emci_lmci	mri_syn_both	mri_real_both	balanced_ACC	1.19239	-3.73404	6.05574

Original6 prespecified comparisons reused; Syn dual PET minus MRI orReal dual PET. Marginal95%5000 fold/class bootstrap CI, uncorrected for multiplicity. No overall DeLong claim and no equivalence inference.

Table S15: Paired comparisons with clinical baselines

task	target	reference	metric	delta_pp	ci_low_pp	ci_high_pp
ad_cn	mri_syn_both	clinical	ACC	-11.9168	-14.907	-9.14853
ad_cn	mri_syn_both	clinical	AUC	-6.17603	-8.82061	-3.9256
ad_cn	mri_syn_both	clinical	SEN	-13.9394	-19.3939	-9.09091
ad_cn	mri_syn_both	clinical	SPE	-10.8167	-14.4309	-7.53899
ad_cn	mri_syn_both	clinical	balanced_ACC	-12.378	-15.7128	-9.32558
ad_cn	mri_syn_both	mri_clinical	ACC	-6.81175	-10.2129	-3.62159
ad_cn	mri_syn_both	mri_clinical	AUC	-5.34224	-7.87614	-3.17673
ad_cn	mri_syn_both	mri_clinical	SEN	-6.06061	-11.5152	-0.606061
ad_cn	mri_syn_both	mri_clinical	SPE	-7.20896	-11.1467	-3.59469
ad_cn	mri_syn_both	mri_clinical	balanced_ACC	-6.63478	-10.198	-3.20788
ad_cn	mri_clinical	mri	ACC	15.3302	12.1373	18.9395
ad_cn	mri_clinical	mri	AUC	12.1443	9.19898	15.4179
ad_cn	mri_clinical	mri	SEN	16.9697	10.9091	23.0303
ad_cn	mri_clinical	mri	SPE	14.4276	10.4995	18.6825
ad_cn	mri_clinical	mri	balanced_ACC	15.6987	12.1825	19.4301
ad_cn	clinical	mri	ACC	20.4352	17.0314	24.046
ad_cn	clinical	mri	AUC	12.9781	9.84708	16.491
ad_cn	clinical	mri	SEN	24.8485	18.7879	31.5152
ad_cn	clinical	mri	SPE	18.0353	14.0911	22.3031
ad_cn	clinical	mri	balanced_ACC	21.4419	17.7596	25.3654
mci_cn	mri_syn_both	clinical	ACC	-29.4197	-32.7935	-26.0459
mci_cn	mri_syn_both	clinical	AUC	-29.4749	-33.432	-25.6737
mci_cn	mri_syn_both	clinical	SEN	-14.2592	-18.3908	-10.1259
mci_cn	mri_syn_both	clinical	SPE	-51.1907	-57.1087	-45.2433
mci_cn	mri_syn_both	clinical	balanced_ACC	-32.7249	-36.388	-29.1387
mci_cn	mri_syn_both	mri_clinical	ACC	-19.7031	-23.2119	-16.3293
mci_cn	mri_syn_both	mri_clinical	AUC	-23.6109	-27.4845	-19.805
mci_cn	mri_syn_both	mri_clinical	SEN	-9.17651	-13.0977	-5.26531
mci_cn	mri_syn_both	mri_clinical	SPE	-34.8185	-41.3675	-28.6029
mci_cn	mri_syn_both	mri_clinical	balanced_ACC	-21.9975	-25.6565	-18.3658
mci_cn	mri_clinical	mri	ACC	17.6788	14.4399	20.9177
mci_cn	mri_clinical	mri	AUC	21.4811	18.0947	24.9968

task	target	reference	metric	delta_pp	ci_low_pp	ci_high_pp
mci_cn	mri_clinical	mri	SEN	8.93718	5.25744	12.6059
mci_cn	mri_clinical	mri	SPE	30.2239	24.0114	36.1681
mci_cn	mri_clinical	mri	balanced_ACC	19.5805	15.9627	23.0734
mci_cn	clinical	mri	ACC	27.3954	24.0216	30.7692
mci_cn	clinical	mri	AUC	27.3451	23.6158	31.1969
mci_cn	clinical	mri	SEN	14.0198	9.88663	18.3782
mci_cn	clinical	mri	SPE	46.5961	40.3837	52.5011
mci_cn	clinical	mri	balanced_ACC	30.308	26.6813	33.8772
emci_lmci	mri_syn_both	clinical	ACC	-33.2593	-37.3973	-28.8994
emci_lmci	mri_syn_both	clinical	AUC	-33.6068	-38.8337	-28.1938
emci_lmci	mri_syn_both	clinical	SEN	-55.6327	-63.701	-46.9656
emci_lmci	mri_syn_both	clinical	SPE	-21.5936	-26.7982	-17.0577
emci_lmci	mri_syn_both	clinical	balanced_ACC	-38.6131	-43.3044	-33.6618
emci_lmci	mri_syn_both	mri_clinical	ACC	-19.4993	-24.0844	-14.9047
emci_lmci	mri_syn_both	mri_clinical	AUC	-23.2724	-28.5706	-17.9808
emci_lmci	mri_syn_both	mri_clinical	SEN	-42.8707	-51.0061	-34.2177
emci_lmci	mri_syn_both	mri_clinical	SPE	-7.35746	-12.6171	-2.13816
emci_lmci	mri_syn_both	mri_clinical	balanced_ACC	-25.1141	-30.0744	-20.0502
emci_lmci	mri_clinical	mri	ACC	20.4188	16.517	24.5458
emci_lmci	mri_clinical	mri	AUC	26.7938	21.7408	31.6749
emci_lmci	mri_clinical	mri	SEN	68.5034	60.4762	75.8912
emci_lmci	mri_clinical	mri	SPE	-4.45906	-8.98026	0.0840643
emci_lmci	mri_clinical	mri	balanced_ACC	32.0222	27.4872	36.569
emci_lmci	clinical	mri	ACC	34.1789	31.1872	37.1706
emci_lmci	clinical	mri	AUC	37.1282	31.5136	42.5461
emci_lmci	clinical	mri	SEN	81.2653	75.2245	86.6395
emci_lmci	clinical	mri	SPE	9.77705	6.28655	13.6111
emci_lmci	clinical	mri	balanced_ACC	45.5212	41.9676	48.8719

12prespecified comparisons x5metrics. Delta target-reference; equal-weight3fold means.5000 paired fold/class bootstrap95% CI, seed20261102. Marginal intervals without simultaneous coverage. Does not estimate repeated-training variation.

Table S16: Paired DeLong comparisons by outer fold

task	fold	target	reference	delta_AUC_pp	p_raw	p_Holm	family
ad_cn	1	mri_syn_both	mri	8.05704	0.00609343	0.109682	existing18
ad_cn	1	mri_syn_both	mri_real_both	0.730838	0.461309	1	existing18
ad_cn	2	mri_syn_both	mri	3.65419	0.113616	1	existing18
ad_cn	2	mri_syn_both	mri_real_both	0.855615	0.40185	1	existing18
ad_cn	3	mri_syn_both	mri	8.69487	0.0202858	0.344858	existing18
ad_cn	3	mri_syn_both	mri_real_both	-3.24032	0.0689018	1	existing18
mci_cn	1	mri_syn_both	mri	-5	0.205433	1	existing18
mci_cn	1	mri_syn_both	mri_real_both	-3.23191	0.299826	1	existing18
mci_cn	2	mri_syn_both	mri	-2.52874	0.425017	1	existing18
mci_cn	2	mri_syn_both	mri_real_both	0.0338066	0.990617	1	existing18
mci_cn	3	mri_syn_both	mri	1.13929	0.767688	1	existing18
mci_cn	3	mri_syn_both	mri_real_both	4.12315	0.203511	1	existing18
emci_lmci	1	mri_syn_both	mri	2	0.73837	1	existing18
emci_lmci	1	mri_syn_both	mri_real_both	4.8125	0.336152	1	existing18
emci_lmci	2	mri_syn_both	mri	6.10119	0.144769	1	existing18
emci_lmci	2	mri_syn_both	mri_real_both	1.70068	0.736135	1	existing18
emci_lmci	3	mri_syn_both	mri	2.46316	0.633676	1	existing18
emci_lmci	3	mri_syn_both	mri_real_both	3.36842	0.383536	1	existing18
ad_cn	1	mri_syn_both	clinical	-4.63458	0.00165732	0.00864088	new36
ad_cn	1	mri_syn_both	mri_clinical	-2.7451	0.037946	0.112295	new36
ad_cn	1	mri_clinical	mri	10.8021	3.40179e-05	0.000408215	new36
ad_cn	1	clinical	mri	12.6916	3.45292e-05	0.000408215	new36
ad_cn	2	mri_syn_both	clinical	-4.02852	0.037572	0.112295	new36
ad_cn	2	mri_syn_both	mri_clinical	-4.0107	0.0374316	0.112295	new36
ad_cn	2	mri_clinical	mri	7.66488	0.00123441	0.00864088	new36
ad_cn	2	clinical	mri	7.68271	0.00126115	0.00864088	new36
ad_cn	3	mri_syn_both	clinical	-9.86499	0.000460643	0.00414579	new36
ad_cn	3	mri_syn_both	mri_clinical	-9.27093	0.000566367	0.00453094	new36
ad_cn	3	mri_clinical	mri	17.9658	5.14555e-08	8.74744e-07	new36
ad_cn	3	clinical	mri	18.5599	1.01156e-07	1.61849e-06	new36
mci_cn	1	mri_syn_both	clinical	-30.4801	6.99058e-18	2.4467e-16	new36
mci_cn	1	mri_syn_both	mri_clinical	-18.1136	1.54431e-07	2.31646e-06	new36

task	fold	target	reference	delta_AUC_pp	p_raw	p_Holm	family
mci_cn	1	mri_clinical	mri	13.1136	1.69917e-07	2.37884e-06	new36
mci_cn	1	clinical	mri	25.4801	3.22727e-15	9.68181e-14	new36
mci_cn	2	mri_syn_both	clinical	-30.6694	4.38095e-19	1.57714e-17	new36
mci_cn	2	mri_syn_both	mri_clinical	-25.0575	2.29151e-14	6.18707e-13	new36
mci_cn	2	mri_clinical	mri	22.5287	5.89174e-13	1.41402e-11	new36
mci_cn	2	clinical	mri	28.1406	1.51617e-16	4.85174e-15	new36
mci_cn	3	mri_syn_both	clinical	-27.2752	2.00445e-14	5.61247e-13	new36
mci_cn	3	mri_syn_both	mri_clinical	-27.6617	3.99106e-15	1.15741e-13	new36
mci_cn	3	mri_clinical	mri	28.801	1.27578e-16	4.33764e-15	new36
mci_cn	3	clinical	mri	28.4145	1.7613e-15	5.46004e-14	new36
emci_lmci	1	mri_syn_both	clinical	-33.0625	1.99145e-12	3.78376e-11	new36
emci_lmci	1	mri_syn_both	mri_clinical	-15.7917	0.00198083	0.00864088	new36
emci_lmci	1	mri_clinical	mri	17.7917	0.000226449	0.00226449	new36
emci_lmci	1	clinical	mri	35.0625	1.37494e-12	3.02487e-11	new36
emci_lmci	2	mri_syn_both	clinical	-33.6947	8.53209e-13	1.96238e-11	new36
emci_lmci	2	mri_syn_both	mri_clinical	-20.0255	4.42156e-07	5.74803e-06	new36
emci_lmci	2	mri_clinical	mri	26.1267	3.80927e-12	6.85669e-11	new36
emci_lmci	2	clinical	mri	39.7959	1.36072e-16	4.49038e-15	new36
emci_lmci	3	mri_syn_both	clinical	-34.0632	1.42116e-12	3.02487e-11	new36
emci_lmci	3	mri_syn_both	mri_clinical	-34	1.62294e-12	3.24588e-11	new36
emci_lmci	3	mri_clinical	mri	36.4632	4.30757e-14	1.11997e-12	new36
emci_lmci	3	clinical	mri	36.5263	4.98084e-14	1.24521e-12	new36

Two-sided paired DeLong only within each outer test fold. Holm separately for original18 and new36 test families. No pooled/combined/mean p-value or three-fold overall significance is asserted. Undefined variance would remain NA and count conservatively in family size.

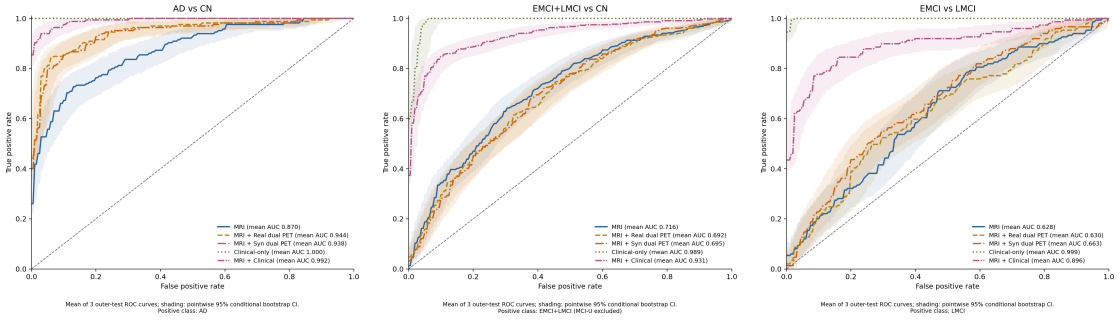


Figure S9: **Cross-sectional classification ROC curves for AD/CN, EMCI+LMCI/CN and EMCI/LMCI (left to right).** Curves average three fold-specific TPRs at a fixed FPR grid; shaded regions are pointwise conditional 95% bootstrap bands, not simultaneous confidence bands. Legend AUC values are equally weighted empirical fold AUC means. Comparisons include MRI, MRI+Real both, MRI+Syn both, Clinical-only and MRI+Clinical. The clinical fields include diagnosis-related scores and historical group-conditioned fills; high clinical AUC is not evidence of independent synthetic-PET benefit.

Table S17: **E1 fixed-weight score-removal sensitivity.** Differences are No-Scores minus Full, except the nonnegative output difference between predictions. This is a distinct E1 Full checkpoint family, not the reconstructed Full used in the generator-comparison tables.

Tracer	Metric	N	Mean Δ	CI lower	CI upper	Holm p
FDG	ROI MAE	1597	3.63611e-05	1.18272e-05	6.01943e-05	0.0035996
FDG	Masked MAE	1597	-1.39166e-05	-3.50725e-05	7.21746e-06	0.22038
FDG	PSNR	1597	-0.00417362	-0.00673816	-0.00160625	–
FDG	SSIM (7)	1597	0.000220733	0.000182378	0.000260921	–
FDG	SSIM (11)	1597	0.000215856	0.000178967	0.000253801	–
FDG	Output difference	1597	0.00101927	0.00100572	0.00103318	–
AV45	ROI MAE	1364	7.87648e-05	4.56388e-06	0.000152402	0.039996
AV45	Masked MAE	1364	0.0001445	9.57843e-05	0.000193458	0.00039996
AV45	PSNR	1364	-0.0201903	-0.0247279	-0.0157227	–
AV45	SSIM (7)	1364	0.000684918	0.000619745	0.000750227	–
AV45	SSIM (11)	1364	0.000554805	0.000491236	0.000619548	–
AV45	Output difference	1364	0.00110469	0.0010868	0.00112321	–

The MRI, tracer context and model weights are fixed; the eight-field patient token is replaced with a frozen Age+Sex+Weight token, not an empty token. No model is retrained. ROI/masked MAE use normalized-intensity units, PSNR is dB, and SSIM is unitless. CIs use 10,000 within-fold paired bootstrap resamples (seed 20261017). ROI and masked MAE are the primary endpoints: paired sign-permutation tests use 10,000 permutations and Holm correction across the two tracers separately within each endpoint. Other intervals are descriptive. The full-reference measurements were reused; whole-cohort Full predictions were not replayed anew. Nonzero output sensitivity and small error changes do not prove independent clinical value.

Table S18: Age- and sex-stratified classification performance: AD vs CN.

Subgroup	Input	N	Negative	Positive	AUC (%) [95% CI]	Balanced ACC (%) [95% CI]
Age $<$ 75	MRI	256	177	79	88.17 [83.22, 92.56]	80.27 [74.84, 85.42]
	MRI+Syn dual PET	256	177	79	93.02 [88.60, 96.90]	87.71 [83.01, 91.96]
	MRI+Real dual PET	256	177	79	93.69 [88.98, 97.46]	90.10 [85.63, 93.98]
Age $\geq$ 75	MRI	214	128	86	84.99 [79.58, 89.90]	76.42 [70.59, 82.07]
	MRI+Syn dual PET	214	128	86	93.83 [90.34, 96.73]	86.33 [81.41, 90.75]
	MRI+Real dual PET	214	128	86	94.55 [91.36, 97.17]	87.74 [83.01, 92.14]
Female	MRI	226	155	71	84.85 [78.40, 90.29]	77.05 [70.86, 82.84]
	MRI+Syn dual PET	226	155	71	94.55 [90.43, 97.73]	88.02 [83.17, 92.52]
	MRI+Real dual PET	226	155	71	94.70 [90.12, 97.88]	88.45 [83.25, 93.04]
Male	MRI	244	150	94	87.54 [82.46, 91.80]	78.59 [73.01, 83.57]
	MRI+Syn dual PET	244	150	94	93.07 [89.24, 96.32]	86.89 [82.16, 91.03]
	MRI+Real dual PET	244	150	94	93.15 [89.20, 96.41]	88.74 [84.24, 92.83]

Metrics are equal means of three within-fold subgroup estimates. Intervals are marginal 95% percentile intervals from 5,000 paired resamples within fold, subgroup and true class, conditional on fixed models and folds; no multiplicity correction was applied. Age and sex are separate subgroup analyses. Positive classes are AD, EMCI+LMCI and LMCI for the respective tasks. Recorded age could not be aligned to the MRI visit because the recovered export lacks visit dates.

Table S19: Age- and sex-stratified classification performance: EMCI+LMCI vs CN.

Subgroup	Input	N	Negative	Positive	AUC (%) [95% CI]	Balanced ACC (%) [95% CI]
Age $<$ 75	MRI	456	177	279	71.56 [66.47, 76.49]	66.28 [62.07, 70.67]
	MRI+Syn dual PET	456	177	279	67.96 [62.99, 72.82]	60.84 [56.65, 65.07]
	MRI+Real dual PET	456	177	279	69.28 [64.47, 74.07]	63.35 [58.88, 67.95]
Age $\geq$ 75	MRI	285	128	157	71.44 [65.36, 77.37]	62.91 [57.85, 68.12]
	MRI+Syn dual PET	285	128	157	71.86 [65.66, 77.68]	64.94 [60.10, 69.99]
	MRI+Real dual PET	285	128	157	70.82 [64.85, 76.86]	64.11 [58.89, 69.50]
Female	MRI	351	155	196	72.83 [67.25, 78.06]	64.58 [59.83, 69.35]
	MRI+Syn dual PET	351	155	196	68.19 [62.27, 73.84]	62.11 [57.49, 66.79]
	MRI+Real dual PET	351	155	196	69.47 [63.90, 74.92]	61.80 [56.79, 66.74]
Male	MRI	390	150	240	70.45 [65.00, 75.74]	65.26 [60.83, 69.83]
	MRI+Syn dual PET	390	150	240	70.79 [65.50, 75.90]	62.89 [58.40, 67.35]
	MRI+Real dual PET	390	150	240	68.85 [63.24, 74.17]	64.64 [59.72, 69.43]

Metrics are equal means of three within-fold subgroup estimates. Intervals are marginal 95% percentile intervals from 5,000 paired resamples within fold, subgroup and true class, conditional on fixed models and folds; no multiplicity correction was applied. Age and sex are separate subgroup analyses. Positive classes are AD, EMCI+LMCI and LMCI for the respective tasks. Recorded age could not be aligned to the MRI visit because the recovered export lacks visit dates.

Table S20: Age- and sex-stratified classification performance: EMCI vs LMCI.

Subgroup	Input	N	Negative	Positive	AUC (%) [95% CI]	Balanced ACC (%) [95% CI]
Age $<$ 75	MRI	279	192	87	63.05 [56.22, 70.16]	54.83 [50.58, 59.32]
	MRI+Syn dual PET	279	192	87	67.22 [60.31, 73.81]	59.69 [54.02, 65.38]
	MRI+Real dual PET	279	192	87	60.90 [53.61, 67.96]	58.22 [53.01, 63.18]
Age $\geq$ 75	MRI	157	95	62	60.34 [50.90, 69.53]	50.57 [44.73, 56.32]
	MRI+Syn dual PET	157	95	62	63.68 [54.37, 72.48]	60.21 [52.28, 67.63]
	MRI+Real dual PET	157	95	62	63.13 [54.27, 71.58]	59.35 [51.65, 67.06]
Female	MRI	196	124	72	62.55 [54.22, 70.81]	55.06 [49.87, 60.73]
	MRI+Syn dual PET	196	124	72	67.23 [59.36, 74.89]	61.64 [54.97, 68.46]
	MRI+Real dual PET	196	124	72	63.21 [55.17, 71.01]	60.65 [54.32, 67.15]
Male	MRI	240	163	77	64.67 [56.97, 72.05]	52.86 [48.50, 57.37]
	MRI+Syn dual PET	240	163	77	65.84 [58.33, 73.24]	58.65 [52.77, 64.92]
	MRI+Real dual PET	240	163	77	63.29 [55.47, 71.07]	57.54 [51.54, 63.57]

Metrics are equal means of three within-fold subgroup estimates. Intervals are marginal 95% percentile intervals from 5,000 paired resamples within fold, subgroup and true class, conditional on fixed models and folds; no multiplicity correction was applied. Age and sex are separate subgroup analyses. Positive classes are AD, EMCI+LMCI and LMCI for the respective tasks. Recorded age could not be aligned to the MRI visit because the recovered export lacks visit dates.